The empirical recurrence for OEIS A305011, recovered from its tail

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Abstract

OEIS A305011 records "Empirical recurrence of order 69" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 2$, so the recurrence is proved for every $n \geq 71$. Its last coefficient is $c_{69} = 1024 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A305011 is "Number of nX4 0..1 arrays with every element unequal to 1, 2, 4, 5 or 7 king-move adjacent elements, with upper left element zero.". It has offset 1 and begins

$$2, 10, 31, 42, 255, 862, 2200, 8807, \ \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (May 23 2018 16:01:49, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 2$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 2$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{69} c_i\, a(n-i),$$

c_1	4	c_{19}	28403762	c_{37}	81188249	c_{55}	22340350
c_2	3	c_{20}	8228759	c_{38}	2012629898	c_{56}	−6454123
c_3	48	c_{21}	−66706420	c_{39}	−265010982	c_{57}	35539621
c_4	−258	c_{22}	−82778038	c_{40}	−440116007	c_{58}	−173787
c_5	−165	c_{23}	−29469002	c_{41}	−1671979104	c_{59}	4135166
c_6	−768	c_{24}	222171648	c_{42}	−15212215	c_{60}	−8094256
c_7	7039	c_{25}	157759060	c_{43}	608572219	c_{61}	−4207396
c_8	3749	c_{26}	102869061	c_{44}	910466026	c_{62}	−2178480
c_9	1672	c_{27}	−505432608	c_{45}	160657681	c_{63}	−254312
c_{10}	−107832	c_{28}	−175403100	c_{46}	−456992400	c_{64}	646848
c_{11}	−46980	c_{29}	−332895161	c_{47}	−287565453	c_{65}	167040
c_{12}	105025	c_{30}	790854205	c_{48}	−217051190	c_{66}	126080
c_{13}	1036367	c_{31}	64733060	c_{49}	229322278	c_{67}	−14080
c_{14}	370074	c_{32}	862691741	c_{50}	25740001	c_{68}	12032
c_{15}	−1679954	c_{33}	−853006996	c_{51}	167068913	c_{69}	1024
c_{16}	−6592643	c_{34}	54981066	c_{52}	−82886501		
c_{17}	−2003461	c_{35}	−1602650274	c_{53}	10884515		
c_{18}	13379733	c_{36}	623424723	c_{54}	−83651553		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 71$, and it is the recurrence OEIS A305011 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 2$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \dots - c_{69}$. Its constant term is $-c_{69} = -1024$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 2)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 24 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 1024.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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